A Digital Twin of India's Pharmaceutical Market

AI-powered simulation of drug adoption across 10,000 synthetic patient personas

Digital Twin Challenge 2026 - Prototype & Code Submission

Problem Statement

- Pharma market research for a new drug launch in India takes 3-6 months and costs Rs. 50L-1Cr+

- India's market is highly heterogeneous: income, city tier, insurance (PM-JAY, CGHS, ESIC), disease burde

- Wrong pricing / wrong segment -> low adoption, poor affordability, wasted launches

- No fast, explainable way to test 'what-if' launch scenarios before committing capital

Healthcare Use Case

- Digital twin of the patient + market population: 10K personas x 5K medicines = 50M simulated interacti

- Predict adoption suitability of a medicine per demographic segment in < 5 seconds

- Pricing optimisation: test price points vs. adoption vs. affordability

- Access & equity: identify segments priced out; impact of insurance-scheme coverage

- Regulatory intelligence: patent / exclusivity horizon & competitor count (FDA Orange Book + NPPA)

Solution & Approach

1. Persona generator - Indian demographic & economic distributions (10 states, 25+ cities)

2. Multi-factor suitability engine - disease match 30%, affordability 25%, pricing 20%,

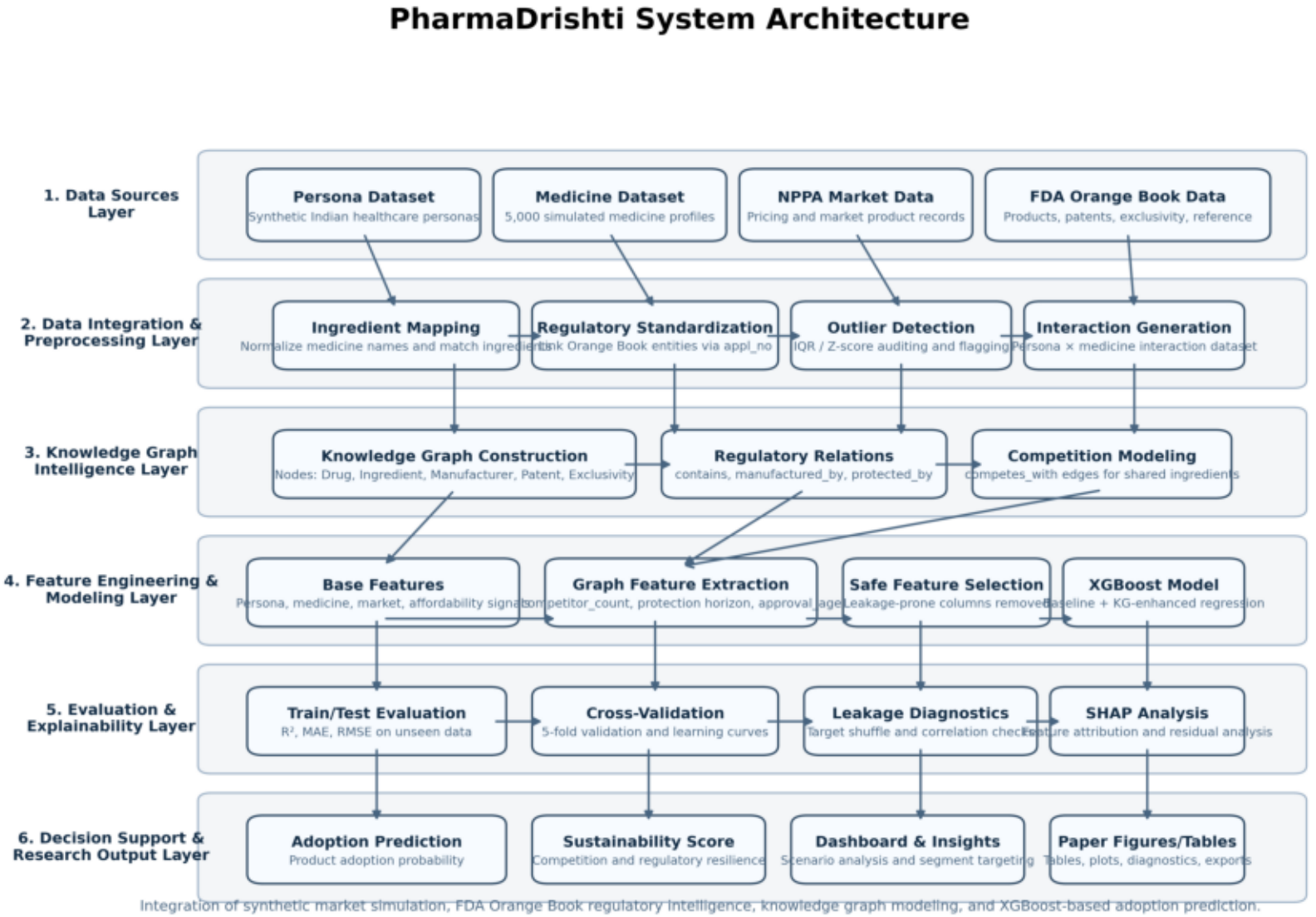
   manufacturing 15%, risk tolerance 10%

3. Regulatory knowledge graph (NetworkX) - Drug, Ingredient, Manufacturer, Patent, Exclusivity

4. XGBoost regressor (baseline + KG-enhanced) with SHAP explanations

5. Streamlit dashboard - scenario simulation, segment ranking, AI insights, CSV export

Architecture



Integration of synthetic market simulation, FDA Orange Book regulatory intelligence, knowledge graph modeling, and XGBoost-based adoption prediction.

Technical Stack

- Language: Python 3.10+

- ML: XGBoost, scikit-learn, SHAP

- Graph: NetworkX knowledge graph + graph feature extraction

- Data: Pandas, NumPy; NPPA price data; FDA Orange Book (products, patents, exclusivity)

- Frontend: Streamlit, Plotly, Matplotlib, Seaborn

- GenAI insights: Google Gemini (optional)

Outcomes & Results

- Validation R^2 > 0.85 on held-out synthetic interactions

- Prediction latency < 5 s per launch scenario across all persona segments

- Leakage-safe pipeline: 5-fold CV, learning curves, target-shuffle test, residual analysis

- Explainable: SHAP feature attribution + plain-language recommendations

- Research time: months -> minutes; est. cost reduction ~95% vs. traditional studies

Limitations & Roadmap

Limitations: synthetic data; no physician-prescription modelling; no post-launch validation yet

Next: 20K+ personas, 20+ disease categories, real-world pilot with pharma partners,

REST API, live NPPA price feed, multi-language dashboard